A counterexample to Oertel's conjecture

Abstract

A conjecture of Oertel, stated for convex bodies by Basu and Oertel, asserts that for every convex body in $\mathbb{R}^n \times \mathbb{R}^d$, $d \geq 1$, of positive mixed-integer volume there is a point of $\mathbb{Z}^n \times \mathbb{R}^d$ such that every closed halfspace containing it holds at least the fraction $2^{-n}(d/(d+1))^d$ of this volume, the sum of the d-dimensional volumes of the fibers over $\mathbb{Z}^n$. We disprove the conjecture at $(n, d) = (12, 5)$ by constructing a rational polytope in $\mathbb{R}^{12} \times \mathbb{R}^5$ such that every point of $\mathbb{Z}^{12} \times \mathbb{R}^5$ lies in a closed halfspace holding at most a fixed fraction $r < (5/6)^5/2^{12}$ of its mixed-integer volume. As in the extremal product of a cube and a simplex, one fiber lies in each nonzero class of $\mathbb{Z}^{12}$ modulo 2 and is separated from all other fibers by tilted halfspaces. The polytope has 756 further fibers in three groups. All three groups add to the total volume, whereas each of the three halfspaces that cover the fiber over the origin holds only one of them. The final strict inequalities are checked in exact rational arithmetic, and the bound on the depth is verified in Lean 4.

1 Introduction

Cutting plane methods for convex minimization measure progress by the volume that a cut removes from the search region. A query point with a fixed fraction of the volume in every closed halfspace containing it therefore guarantees progress whatever the direction of the cut, and in $\mathbb{R}^d$ such a point always exists. Grünbaum [5, Theorem 2 and Section 3] proved that every closed halfspace containing the centroid of a convex body in $\mathbb{R}^d$, $d \geq 1$, holds at least the fraction $\Gamma_d = (d/(d+1))^d$ of its volume, and this fraction is best possible [5, Section 4]. In convex mixed-integer optimization the volume becomes a sum of volumes of fibers, and Basu and Oertel [1, Section 2 and Theorem 2.6] bound the number of first-order oracle calls of a centerpoint cutting plane method through the fraction guaranteed at the query point. We show that the fraction conjectured by Oertel fails for 12 integer and 5 continuous variables.

Let $n \geq 0$ and $d \geq 1$ be integers. We write points of $\mathbb{R}^n \times \mathbb{R}^d$ as $(\mathbf{z}, \mathbf{x})$ and call $\mathbb{Z}^n \times \mathbb{R}^d$ the *mixed-integer lattice*. The *fiber* of a set $D \subseteq \mathbb{R}^n \times \mathbb{R}^d$ over $\mathbf{z} \in \mathbb{Z}^n$ is $D_{\mathbf{z}} = \{\mathbf{x} \in \mathbb{R}^d : (\mathbf{z}, \mathbf{x}) \in D\}$. The *mixed-integer volume*, or *mass*, of a set D with Lebesgue measurable fibers is the sum $\mu(D) = \sum_{\mathbf{z} \in \mathbb{Z}^n} \mathrm{vol}(D_{\mathbf{z}})$ of their d-dimensional volumes. A *halfspace* is a set $\{F \geq b\} = \{(\mathbf{z}, \mathbf{x}) : F(\mathbf{z}, \mathbf{x}) \geq b\}$, where F is a nonzero linear functional on $\mathbb{R}^n \times \mathbb{R}^d$ and $b \in \mathbb{R}$. Let $C \subset \mathbb{R}^n \times \mathbb{R}^d$ be a convex body, that is, a compact convex set with nonempty interior, and let $\mu(C) > 0$. The *depth* of a point $(\mathbf{z}, \mathbf{x})$ of the mixed-integer lattice is the infimum, over the halfspaces H containing the point, of the fraction of the mass of C that H holds (Figure 1(a)),

$$\mathrm{depth}_C(\mathbf{z}, \mathbf{x}) = \inf_{H \ni (\mathbf{z}, \mathbf{x})} \frac{\mu(C \cap H)}{\mu(C)}.$$

Compactness leaves finitely many nonempty fibers, so $\mu(C) < \infty$, and restricting H to the halfspaces with the point on their boundary, as in [1, (2)], does not change the infimum.

Two extreme cases suggest the fraction to expect. For $n = 0$ the mass is the volume, and Grünbaum's theorem gives depth at least Γ_d at the centroid. For $d = 0$ the mass counts the integer points of C, and the general bound $1/(2^n(d+1))$ of Oertel [6, Theorem 4.1.19] and Basu and Oertel [1, Corollary 3.4] gives 2^{-n}. The product of the unit cube and a d-simplex combines the two factors [6, Section 4.1, after Conjecture 4.1.20], since its largest depth is exactly $2^{-n}\Gamma_d$ (Section 2). Oertel [6, Conjecture 4.1.20], for closed convex sets, and Basu and Oertel [1, Conjecture 3.5], for

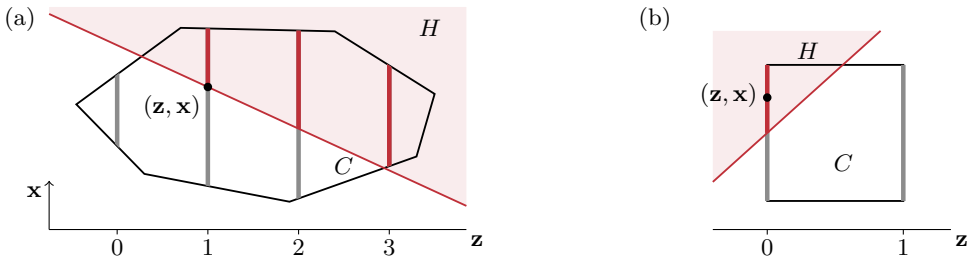


Figure 1: Depth for $n = d = 1$. (a) The fibers of C are its vertical chords over the integers, and its mass is their total length. The halfspace H contains the point $(\mathbf{z}, \mathbf{x})$ and holds the red parts of the fibers. (b) In the product $C = [0,1] \times [0,1]$, the halfspace through $(0, 1/2)$ is tilted to miss the fiber over 1. It contains every point $(0, \mathbf{x})$ with $\mathbf{x} \geq 1/2$ and holds $1/4 = 2^{-1}\Gamma_1$ of the mass.

convex bodies, conjectured that this fraction is always reached, so that every convex body of positive mass would have a point of $\mathbb{Z}^n \times \mathbb{R}^d$ of depth at least $2^{-n}\Gamma_d$. We disprove this at $(n, d) = (12, 5)$, where the fraction is $(5/6)^5/2^{12}$.

Theorem 1.1. *There is a rational polytope $C \subset \mathbb{R}^{12} \times \mathbb{R}^5$ with nonempty interior and $\mu(C) > 0$ such that $\sup_{(\mathbf{z},\mathbf{x}) \in \mathbb{Z}^{12} \times \mathbb{R}^5} \operatorname{depth}_C(\mathbf{z}, \mathbf{x}) < (5/6)^5/2^{12}$.*

The proof gives a rational number $r < (5/6)^5/2^{12}$ and, for every point of $\mathbb{Z}^{12} \times \mathbb{R}^5$, a halfspace containing the point that holds at most the fraction r of the mass. The masses are compared in exact rational arithmetic (Section 6), and Lean 4 verifies the bound on the depth (Appendix A). The margin is small, with $10^5 r = 9.811464096\ldots$ against $10^5(5/6)^5/2^{12} = 9.811464160\ldots$.

For $d \geq 2$ the general bound $1/(2^n(d+1))$ lies below the conjectured fraction, and the following results go beyond it under additional structure. For large projections onto $\mathbb{R}^n$, Basu and Oertel [1, Theorem 3.6] proved the conjectured bound when the projection has lattice width above a threshold exponential in n, Cristi and Salas [4, Theorem 4.5] proved it when the projection contains a Euclidean ball of radius at least $\alpha d^2 n^{3/2}$ for a universal constant α, and Cheng and Basu [2, Corollary 2.5] showed that a translate of the ℓ_∞ ball of radius $k \geq 3e(n+d)$ in the projection gives depth at least $1/(2e)$. For few fibers, Cheng and Basu [3, Theorem 1] proved the fraction $2/9 = \Gamma_2/2$ for compact convex sets in $\mathbb{R} \times \mathbb{R}^2$ with exactly three slices of positive area. The polytope of Theorem 1.1 has 4852 nonempty fibers, and its projection onto $\mathbb{R}^{12}$ lies in $[-1, 1]^{12}$.

2 Overview of the construction

The product example. Let $S \subset \mathbb{R}^d$ be a d-simplex and $C = [0,1]^n \times S$, whose nonempty fibers are 2^n copies of S. A point $\mathbf{x} \in S$ has a barycentric coordinate at least $1/(d+1)$, and the set where this coordinate is at least $1/(d+1)$ is a homothetic copy of S with ratio $d/(d+1)$. We call such a set a *cap* of S. It has volume $\Gamma_d \operatorname{vol}(S)$ and is cut from S by a halfspace of $\mathbb{R}^d$, which extends to a halfspace of $\mathbb{R}^n \times \mathbb{R}^d$ that misses all other fibers. Indeed, a vertex $\mathbf{z}$ of the cube is the unique maximizer over $\{0,1\}^n$ of some linear functional on $\mathbb{R}^n$, and adding a large multiple of this functional tilts the halfspace away from the other fibers without changing it over $\mathbf{z}$ (Figure 1(b)). Hence every point of $\mathbb{Z}^n \times \mathbb{R}^d$ has depth at most $2^{-n}\Gamma_d$ in C. Conversely, a halfspace containing a point $(\mathbf{z}, \mathbf{x})$ meets the fiber over $\mathbf{z}$ in a halfspace of $\mathbb{R}^d$ containing $\mathbf{x}$ or in all of $\mathbb{R}^d$. Grünbaum's theorem then gives the centroid of a fiber depth at least $2^{-n}\Gamma_d$, so the largest depth in C is $2^{-n}\Gamma_d$.

The second argument limits any counterexample. If a convex body has at most 2^n fibers of positive volume, then a largest fiber carries at least the fraction 2^{-n} of the mass, and its centroid has depth at least $2^{-n}\Gamma_d$. A counterexample therefore has more than 2^n such fibers, and two of them

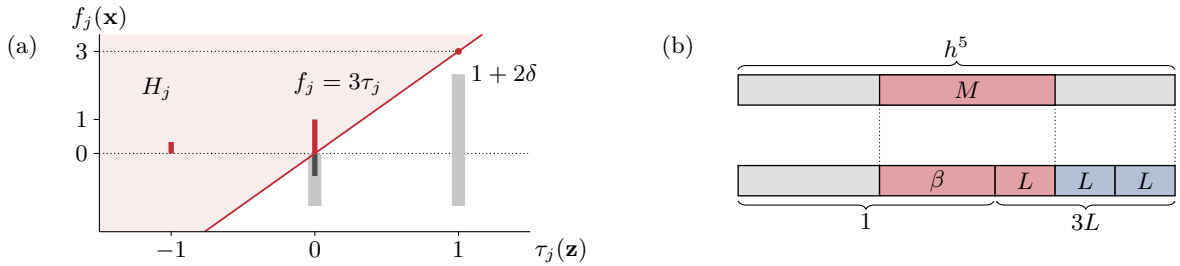


Figure 2: (a) The cut H_j in the coordinates $(\tau_j(\mathbf{z}), f_j(\mathbf{x}))$, where it is the halfplane above the line $f_j = 3\tau_j$. Each fiber is a vertical segment over the integer $\tau_j(\mathbf{z})$, drawn with the range of f_j at the parameters of Section 6. These are the light fibers of the jth pool over -1, the origin fiber (narrow) and all other fibers (wide) over 0, and heavy fibers over 1 and, not shown, over larger integers. Red marks the parts of positive volume that H_j holds. (b) Volumes to scale. Top: a heavy fiber of volume h^5 with a cap of volume M. Bottom: the origin fiber, of volume 1, and the three pools. Red marks what H_1 holds.

lie over points $\mathbf{z} \neq \mathbf{z}'$ in the same class of $(\mathbb{Z}/2\mathbb{Z})^n$. By convexity the body then has a nonempty fiber over the integer point $(\mathbf{z} + \mathbf{z}')/2$ as well. In the polytope of Theorem 1.1 this midpoint is always the origin.

The mass count. We call a halfspace H a *cut* and $\mu(C \cap H)$ its mass. The proof gives a polytope C of mass W and a number M such that every point of the mixed-integer lattice lies in a cut of mass at most M and thus has depth at most M/W. The polytope is the convex hull of polytopes $\{\mathbf{z}\} \times A_{\mathbf{z}}$ over 4852 integer points $\mathbf{z}$, which we call *bases*, and its nonempty fibers are exactly the sets $A_{\mathbf{z}}$. We use units in which the fiber over the origin has volume 1.

One *heavy* base lies in each of the 4095 nonzero classes of $(\mathbb{Z}/2\mathbb{Z})^{12}$, and we treat it as in the product example. A heavy base is a vertex of the convex hull of the bases, so a halfspace of $\mathbb{R}^5$ extends to a cut that holds the same part of its fiber and misses all other fibers. The heavy fibers have volume at most h^5 for a parameter h, and each of their points lies in such a part of volume at most $M = \Gamma_5 h^5$, the volume of a cap of a simplex of volume h^5. With this M, the inequality $M/W < \Gamma_5/2^{12}$ reads

$$W > 4096\, h^5. \tag{1}$$

Equality would hold for 4096 fibers of volume h^5. The heavy fibers contribute at most $4095h^5$ and the origin fiber contributes $1 < h^5$, so (1) needs further fibers.

Light fibers and the origin cuts. The further fibers lie over 756 *light* bases. Each light base is the negative $-\mathbf{z}$ of a heavy base $\mathbf{z}$, and the light bases form three *pools* of 252 whose fibers have total volume L per pool, so W gains $3L$. Tilting cannot isolate the origin fiber from a light fiber and its heavy partner at once, because a linear functional on $\mathbb{R}^{12}$ that is smaller at $\mathbf{z}$ than at $\mathbf{0}$ is larger at $-\mathbf{z}$. We turn this to our advantage with three cuts

$$H_j = \{(\mathbf{z}, \mathbf{x}) : f_j(\mathbf{x}) \geq 3\tau_j(\mathbf{z})\}, \qquad j = 1, 2, 3,$$

where f_1, f_2, f_3 are linear functionals on $\mathbb{R}^5$ with $f_1 + f_2 + f_3 = 0$ and τ_1, τ_2, τ_3 are linear forms on $\mathbb{R}^{12}$ with integer coefficients. Over the origin, H_j holds the part $\{f_j \geq 0\}$ of the origin fiber, of volume β, and these three parts cover the origin fiber because the f_j sum to zero. Over another base the integer $\tau_j(\mathbf{z})$ moves the threshold in steps of 3, which is more than $|f_j|$ on any fiber. Hence H_j contains the fiber if $\tau_j(\mathbf{z}) \leq -1$ and misses it if $\tau_j(\mathbf{z}) \geq 1$ (Figure 2(a)). The bases are chosen so that $\tau_j = -1$ exactly on the jth pool, and the fibers so that every fiber with $\tau_j(\mathbf{z}) = 0$ other

j	$\mathbf{s}$	$\tau(\mathbf{b}_J, \mathbf{s})$	$\sigma(\mathbf{b}_J, \mathbf{s})$	Paired	Near
1	$\mathbf{e}_1$	$(u,\ u-1,\ u-1)$	$3u-2$	$u=1$	$u=0$
2	$\mathbf{e}_2$	$(u-1,\ u,\ u-1)$	$3u-2$	$u=1$	$u=0$
3	$\mathbf{e}_1+\mathbf{e}_2$	$(u-1,\ u-1,\ u-2)$	$3u-4$	$u=1$	$u=2$

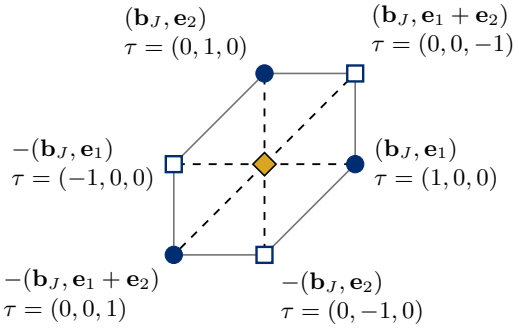


Figure 3: Left: the values of τ and σ at the candidates $(\mathbf{b}_J, \mathbf{s})$ of type j, with $u = u(\mathbf{b}_J)$, and the values of u that give paired and near bases. Right: the seven bases obtained from $J = \{1, 5, 6\}$ and the origin, plotted by selector $\mathbf{s}$. Circles are paired bases, squares are light bases, and the diamond marks the selector $\mathbf{0}$ of the origin, with $\tau = (0, 0, 0)$, and of the regular base $(\mathbf{1}_J, \mathbf{0})$, with $\tau = (3, 3, 3)$. Dashed segments join antipodal bases. The hexagon is the convex hull of the selectors of all bases.

than the origin fiber lies in $\{f_j \le 0\}$, where H_j holds none of its volume. The cut H_j thus has mass $\beta + L$. Each of the three cuts pays for one pool, while all three pools count in W (Figure 2(b)). At our parameters the mass $\beta + L \approx 0.68314$ of each cut lies just below $M \approx 0.68315$, and the origin fiber and the three pools have total volume $1 + 3L \approx 1.70017$, just above $h^5 \approx 1.69988$.

Two requirements shape the bases of Section 3 and the fibers of Section 4. First, the convex hull must not create new integer fibers or enlarge the given ones. The bases turn out to be the only integer points of their convex hull (Lemma 5.1), and we place each fiber inside a dilation of the origin fiber whose factor is affine in $\mathbf{z}$ and equals 1 at the origin, so that the midpoints of antipodal fibers stay in the origin fiber (Lemma 5.2). Second, the heavy partner of a light base of the jth pool has $\tau_j = 1$ and $\tau_i = 0$ for $i \ne j$, so its fiber must lie in $\{f_i \le 0\}$ for both $i \ne j$. This fiber is a simplex with a thin slab attached, and the parts that halfspaces cut from it need a separate computation (Lemma 6.3).

3 The integer bases

We write $\mathbf{z} = (\mathbf{y}, \mathbf{s}) \in \mathbb{R}^{10} \times \mathbb{R}^2$, call $\mathbf{s}$ the *selector*, and put $u = \sum_i y_i$. Let

$$\tau(\mathbf{z}) = (u - s_2,\ u - s_1,\ u - s_1 - s_2), \qquad \sigma(\mathbf{z}) = \tau_1(\mathbf{z}) + \tau_2(\mathbf{z}) + \tau_3(\mathbf{z}) = 3u - 2s_1 - 2s_2.$$

The entries of τ are the thresholds of the origin cuts, and σ will set the dilation factors of the fibers. A light base of the jth pool needs $\tau_j = -1$ and $\tau_i = 0$ for $i \ne j$, so its heavy partner has τ equal to a unit vector, and we want many such points in distinct classes modulo 2.

Let $\mathbf{e}_1, \mathbf{e}_2$ be the unit vectors of $\mathbb{R}^2$, and let $c_i = -1$ for $i \le 4$ and $c_i = 1$ for $5 \le i \le 10$. For $J \subseteq \{1, \dots, 10\}$, let $\mathbf{1}_J$ be the indicator vector of J, and let $\mathbf{b}_J$ have entries c_i on J and 0 off J. The *candidates* are the points $(\mathbf{1}_J, \mathbf{0})$ with $J \ne \emptyset$ and the points $(\mathbf{b}_J, \mathbf{s})$ with $\mathbf{s} \in \{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_1 + \mathbf{e}_2\}$, which we say have *type* $1, 2, 3$, respectively. Modulo 2 the vector $\mathbf{b}_J$ reduces to $\mathbf{1}_J$ and the selectors $\mathbf{0}, \mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_1 + \mathbf{e}_2$ are distinct, so the 4095 candidates represent each nonzero class of $(\mathbb{Z}/2\mathbb{Z})^{12}$ exactly once. The signs c_i do not alter the class, but they move u. Since $u(\mathbf{b}_J) = |J \triangle \{1, 2, 3, 4\}| - 4$ and the map $J \mapsto J \triangle \{1, 2, 3, 4\}$ permutes the subsets of $\{1, \dots, 10\}$, exactly $\binom{10}{k+4}$ sets J have $u(\mathbf{b}_J) = k$. In particular $u(\mathbf{b}_J) = 1$ for $\binom{10}{5} = 252$ sets, whereas $u(\mathbf{1}_J) = 1$ for only 10. Figure 3 lists τ and σ at the candidates of the three types.

At the candidates $(\mathbf{b}_J, \mathbf{s})$ the value of σ is not divisible by 3, and $\sigma(\mathbf{1}_J, \mathbf{0}) = 3|J|$, so σ is nonzero at every candidate. A *heavy base* is a candidate multiplied by the sign of σ. Heavy bases thus have $\sigma > 0$ and still represent each nonzero class exactly once. After this change of sign, a heavy base of type j with u as in the column "Paired" has $\tau_j = 1$ and zeros elsewhere, one with u as in the column "Near" has $\tau_j = 0$ and ones elsewhere, and every other value of u gives three positive entries, as do the bases $(\mathbf{1}_J, \mathbf{0})$ with $\tau = (|J|, |J|, |J|)$. We call these heavy bases *paired* of type j, *near* of type j and *regular*, respectively. For each paired base $\mathbf{z}$ of type j, the point $-\mathbf{z}$ is a *light* base of type j, with $\tau_j = -1$ and zeros elsewhere, and these are the bases of the jth pool. The set $\mathcal{Z}$ of *bases* consists of $\mathbf{0}$ and all heavy and light bases. Hence σ equals 1 at paired, -1 at light and 2 at near bases, and is at least 3 at regular bases.

Each type has 252 paired bases and $\binom{10}{4} = \binom{10}{6} = 210$ near bases, so there are 756 paired, 630 near and $4095 - 756 - 630 = 2709$ regular bases. A light base lies in the class of its partner and differs from it, so $\mathcal{Z}$ has $1 + 4095 + 756 = 4852$ points. Figure 3 also shows the bases obtained from $J = \{1, 5, 6\}$, where $\mathbf{b}_J$ has entries $-1, 1, 1$ in positions $1, 5, 6$ and $u = 1$. The candidates of types 1 and 2 are paired bases as they stand, whereas $\tau(\mathbf{b}_J, \mathbf{e}_1 + \mathbf{e}_2) = (0, 0, -1)$ has negative sum, so the paired base of type 3 is $-(\mathbf{b}_J, \mathbf{e}_1 + \mathbf{e}_2)$ and its light partner is the candidate itself.

4 The fibers

Let $B = \{\mathbf{x} \in \mathbb{R}^6 : x_i \geq 0,\ \sum_i x_i = 1\}$ be the standard simplex with vertices $\mathbf{v}_1, \ldots, \mathbf{v}_6$. We identify the hyperplane $\sum_i x_i = 1$ with $\mathbb{R}^5$ by dropping x_6 and subtracting $1/6$ from the other coordinates. The centroid of B is then the origin of $\mathbb{R}^5$, scalar multiples of subsets are dilations about the centroid, and $x_1, \ldots, x_6$ are affine functions on $\mathbb{R}^5$ with sum 1. From now on vol and μ are measured in units of the Lebesgue volume of B, which does not change the ratios in Theorem 1.1. The parameters $a, \delta, \varepsilon, h$ introduced below are fixed in Section 6.

The origin cuts. Fix a rational number a with $1/2 < a < 1$. For $j = 1, 2, 3$, with these three indices taken cyclically, put $f_j = x_j - a x_{j+1} - (1 - a) x_{j+2}$. Each f_j vanishes at the centroid and equals 1 at $\mathbf{v}_j$, so it is a nonzero linear functional on $\mathbb{R}^5$. Moreover $|f_j| \leq 1$ on B and $f_1 + f_2 + f_3 = 0$. The cyclic permutation of $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ is a volume-preserving linear symmetry of B that carries f_j to f_{j+1}, so we describe the sets below for $j = 1$ and obtain the others by rotation. The *origin cuts* are the halfspaces $H_j = \{(\mathbf{z}, \mathbf{x}) \in \mathbb{R}^{12} \times \mathbb{R}^5 : f_j(\mathbf{x}) \geq 3\tau_j(\mathbf{z})\}$ for $j = 1, 2, 3$.

Fix also a rational number δ with $0 < \delta < 1$. The fiber $A_{\mathbf{z}}$ over a base $\mathbf{z}$ will be a subset of $(1 + \delta\sigma(\mathbf{z}))B$, and every heavy fiber will moreover lie in $(1 + 2\delta)B$, where $|f_j| \leq 1 + 2\delta < 3$, so that H_j misses it when $\tau_j(\mathbf{z}) \geq 1$. When $\tau_j(\mathbf{z}) = 0$ and $\mathbf{z} \neq \mathbf{0}$, we ask that $A_{\mathbf{z}} \subseteq \{f_j \leq 0\}$. Then H_j meets $A_{\mathbf{z}}$ inside the hyperplane $\{f_j = 0\}$, which is null. By the values of τ in Section 3, a paired or light fiber of type j should therefore lie in $\{f_i \leq 0\}$ for both $i \neq j$ and a near fiber of type j should lie in $\{f_j \leq 0\}$, while a regular fiber needs no sign condition. Dilations about the origin preserve these conditions because the f_j are linear.

Volumes on the active face. Each f_j is a homogeneous linear form in (x_1, x_2, x_3). Let the *trace* of a subset of B be its intersection with the *active face* $\mathrm{conv}(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$. A subset of B defined by inequalities $f_j \geq 0$ or $f_j \leq 0$ is then the join of its trace with $\mathrm{conv}(\mathbf{v}_4, \mathbf{v}_5, \mathbf{v}_6)$, that is, the union of the segments between these two sets. Indeed, normalizing the first three coordinates by their sum when it is positive preserves the inequalities, and points with that sum zero belong to the latter triangle. The simplex spanned by three points of the active face and $\mathbf{v}_4, \mathbf{v}_5, \mathbf{v}_6$ has volume equal to the absolute determinant of the 3×3 matrix of barycentric coordinates of these points, because the affine map carrying the vertices of B to the vertices of this simplex acts on barycentric coordinates

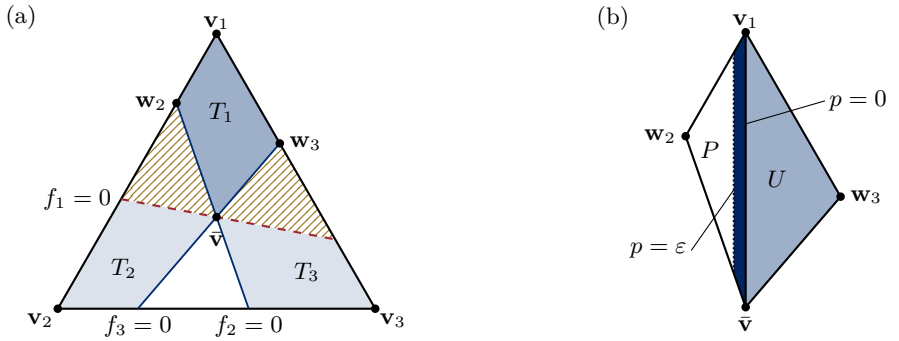


Figure 4: Traces on the active face $\mathrm{conv}(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$, drawn at $a = 663/1000$. (a) The lines $f_j = 0$ meet at the centroid $\bar{\mathbf{v}}$ of the face and bound the shaded traces of T_1, T_2, T_3. The trace of $B \cap \{f_1 \geq 0\}$ consists of T_1 and the two hatched triangles. (b) The trace of T_1, enlarged. The dark strip is the trace of the slab $P \cap \{p \leq \varepsilon\}$, drawn at $p = 1/5$ for the true $\varepsilon = 13/20000$, and with U it forms the trace of K_1. The slab is not a join, and its volume is not the area of its trace.

as the direct sum of this matrix and the identity. We call this the determinant formula. The same determinant is the area of the triangle spanned by the three points as a fraction of the area of the active face, so for such simplices Figure 4 shows volumes as areas. On the active face, f_1 is positive at $\mathbf{v}_1$, negative at $\mathbf{v}_2$ and $\mathbf{v}_3$, and zero at $(a\mathbf{v}_1 + \mathbf{v}_2)/(1 + a)$ and $((1 - a)\mathbf{v}_1 + \mathbf{v}_3)/(2 - a)$. So $B \cap \{f_1 \geq 0\}$ is a simplex, and the determinant formula gives its volume $\beta = 1/[(1 + a)(2 - a)]$.

Paired and light fibers. The sign conditions confine a paired or light fiber of type j to a dilation of $T_j = B \cap \{f_{j+1} \leq 0,\ f_{j+2} \leq 0\}$, on which $f_j = -f_{j+1} - f_{j+2} \geq 0$. Among the vertices of the active face, f_2 is positive only at $\mathbf{v}_2$ and f_3 only at $\mathbf{v}_3$. The lines $f_2 = 0$ and $f_3 = 0$ meet at the centroid $\bar{\mathbf{v}} = (\mathbf{v}_1 + \mathbf{v}_2 + \mathbf{v}_3)/3$ of the face. They meet the edges from $\mathbf{v}_1$ to $\mathbf{v}_2$ and from $\mathbf{v}_1$ to $\mathbf{v}_3$ at $\mathbf{w}_2 = (\mathbf{v}_1 + (1 - a)\mathbf{v}_2)/(2 - a)$ and $\mathbf{w}_3 = (\mathbf{v}_1 + a\mathbf{v}_3)/(1 + a)$, respectively. The edge from $\mathbf{v}_2$ to $\mathbf{v}_3$ has $f_1 < 0$ throughout, so it is disjoint from T_1. The trace of T_1 is therefore the quadrilateral with vertices $\mathbf{v}_1, \mathbf{w}_2, \bar{\mathbf{v}}, \mathbf{w}_3$ in Figure 4(a), whose diagonal from $\mathbf{v}_1$ to $\bar{\mathbf{v}}$ lies on the line $x_2 = x_3$. Consequently $T_1 = U \cup P$ with $U = \mathrm{conv}(\mathbf{v}_1, \bar{\mathbf{v}}, \mathbf{w}_3, \mathbf{v}_4, \mathbf{v}_5, \mathbf{v}_6)$ and $P = \mathrm{conv}(\mathbf{v}_1, \bar{\mathbf{v}}, \mathbf{w}_2, \mathbf{v}_4, \mathbf{v}_5, \mathbf{v}_6)$. These simplices lie on opposite sides of the hyperplane $x_2 = x_3$ and meet in a common facet, which is a null set. By the determinant formula, U has volume $\theta = a/[3(1 + a)]$ and P has volume $\phi = (1 - a)/[3(2 - a)]$.

A light fiber of type j can be all of $(1 - \delta)T_j$, because the only cut we use on it is H_j, which contains it. In a paired fiber, each point must lie in a part of volume at most M cut off by a halfspace of $\mathbb{R}^5$. The caps of the simplex $(1 + \delta)U$ have this property, but at our parameters its volume is below h^5 by more than $3 \cdot 10^{-3}$, and 756 such deficits would exceed $3L$. We therefore add to U a thin slab of P along the common facet. It restores all but about $3 \cdot 10^{-7}$ of the volume h^5, and the parts cut off by the same halfspaces can still be computed (Lemma 6.3).

To define the slab, let $\nu_0, \ldots, \nu_5$ be the barycentric coordinates of U in the listed vertex order and put $q_i = \nu_i(\mathbf{w}_2)$. Comparing active coordinates gives $q_1 = 3(1-a)/(2-a)$, $q_2 = -(1-a^2)/[a(2-a)]$, $q_0 = 1 - q_1 - q_2$ and $q_3 = q_4 = q_5 = 0$. Since $1/2 < a < 1$, we have $0 < q_1 < 1$, $q_2 < 0$ and $0 < q_1 + q_2 = q_1(2a-1)/(3a) < 1$, so $0 < q_0 < 1$. Fix a rational number ε with $0 < \varepsilon < 1/1000$ and put $K_1 = T_1 \cap \{\nu_2 \geq \varepsilon q_2\}$. On U we have $\nu_2 \geq 0 > \varepsilon q_2$. On P, let p be the barycentric coordinate at $\mathbf{w}_2$ and, for $i \neq 2$, let p_i be the barycentric coordinate at the vertex of U with index i. Then $\nu_i = p_i + q_i p$ for $i \neq 2$ and $\nu_2 = q_2 p$, since both sides are affine and agree at the vertices of P. Therefore K_1 is the union of U and the slab $P \cap \{p \leq \varepsilon\}$. The set $P \cap \{p \geq \varepsilon\}$ is a homothetic copy of P with ratio $1 - \varepsilon$, so $\mathrm{vol}(K_1) = V := \theta + \phi[1 - (1 - \varepsilon)^5]$. Rotation gives K_2, K_3.

Near and regular fibers. A near fiber of type j has dilation factor $1+2\delta$ and must lie in $\{f_j \le 0\}$. Fix a rational number $h > 0$ with $2h < 1+2\delta$ and put $\eta = (h/(1+2\delta))^5$. Then $\eta < 1/32$, and $a > 1/2$ gives $a/(1+a) > 1/3$, so $\eta < a/(1+a)$. Let $N_j = \operatorname{conv}\big(\eta\mathbf{v}_j + (1-\eta)\mathbf{v}_{j+1}, \mathbf{v}_{j+1}, \mathbf{v}_{j+2}, \mathbf{v}_4, \mathbf{v}_5, \mathbf{v}_6\big) \subseteq B$. At these vertices f_j takes the values $\eta - a(1-\eta)$, $-a$, $a-1$, 0, 0, 0, which are nonpositive because $\eta < a/(1+a)$. Hence $f_j \le 0$ on N_j, and $\operatorname{vol}(N_j) = \eta$ by the determinant formula, so the simplex $(1+2\delta)N_j$ has volume h^5. A regular fiber is the simplex hB. The table lists the fibers $A_{\mathbf{z}}$.

Base $\mathbf{z}$	Number	$\sigma(\mathbf{z})$	$\tau(\mathbf{z})$	Fiber $A_{\mathbf{z}}$	$\operatorname{vol}(A_{\mathbf{z}})$
$\mathbf{0}$	1	0	$(0,0,0)$	B	1
paired, type j	252 per type	1	$\tau_j = 1$, others 0	$(1+\delta)K_j$	$(1+\delta)^5 V$
light, type j	252 per type	-1	$\tau_j = -1$, others 0	$(1-\delta)T_j$	$(1-\delta)^5(\theta+\phi)$
near, type j	210 per type	2	$\tau_j = 0$, others 1	$(1+2\delta)N_j$	h^5
regular	2709	≥ 3	all entries ≥ 1	hB	h^5

Every fiber lies in $(1+\delta\sigma(\mathbf{z}))B$, where for regular bases we use $h < 1+2\delta < 1+\delta\sigma(\mathbf{z})$ and $\mathbf{0} \in B$, and every heavy fiber lies in $(1+2\delta)B$. The sign conditions hold by construction. Let $C = \operatorname{conv}\big(\bigcup_{\mathbf{z}\in\mathcal{Z}} \{\mathbf{z}\} \times A_{\mathbf{z}}\big)$. The light fibers of one type have total volume $L = 252(\theta+\phi)(1-\delta)^5$, and all fibers have total volume $W = 3339h^5 + 756(1+\delta)^5 V + 1 + 3L$. The comparison with the target (1) therefore reads

$$W - 4096h^5 = 3L - (h^5 - 1) - 756\big[h^5 - (1+\delta)^5 V\big]. \tag{2}$$

5 The integer fibers of the polytope

We first show that a nonzero integer point of $\operatorname{conv}(\mathcal{Z})$ has only the trivial representation.

Lemma 5.1. *If $\mathbf{z} \in \mathbb{Z}^{12} \setminus \{\mathbf{0}\}$ is a convex combination $\sum_m \lambda_m \mathbf{z}^m$ of points $\mathbf{z}^m \in \mathcal{Z}$ with all $\lambda_m > 0$, then $\mathbf{z}^m = \mathbf{z}$ for every m.*

Proof. All entries of all bases lie in $\{-1, 0, 1\}$, so $\mathbf{z} = (\mathbf{y}, \mathbf{s})$ has the same property. If a coordinate of $\mathbf{z}$ equals ± 1, or if the contributors $\mathbf{z}^m = (\mathbf{y}^m, \mathbf{s}^m)$ take only two consecutive integer values in that coordinate, then integrality forces all contributors to agree in it. We call this saturation.

Suppose first that $\mathbf{s} \neq \mathbf{0}$, say $s_k = \pm 1$. Saturation gives $s_k^m = s_k$ for all m. Every nonzero selector of a base is $\pm\mathbf{e}_1$, $\pm\mathbf{e}_2$ or $\pm(\mathbf{e}_1 + \mathbf{e}_2)$, so the other selector coordinate of each contributor lies in $\{0, s_k\}$, and saturation gives $\mathbf{s}^m = \mathbf{s}$. Let $\omega = \pm 1$ be the sign of the nonzero entries of $\mathbf{s}$. By construction, a base with this selector is $(\omega\mathbf{b}_J, \mathbf{s})$ for some J, so $y_i^m \in \{0, \omega c_i\}$ for each i, and saturation gives $\mathbf{y}^m = \mathbf{y}$.

Now suppose that $\mathbf{s} = \mathbf{0}$. Then $y_i = \pm 1$ for some i, and saturation gives $y_i^m = y_i$ for all m. A contributor with a nonzero selector of sign ω has $y_i = \omega c_i$, so $c_i y_i (s_1^m + s_2^m) = \omega(s_1^m + s_2^m) > 0$, while a contributor with selector $\mathbf{0}$ gives 0. The weighted sum of these numbers is $c_i y_i (s_1 + s_2) = 0$, so all contributors have selector $\mathbf{0}$. They are $\mathbf{0}$ or points $(\mathbf{1}_J, \mathbf{0})$, so $\mathbf{y}^m \in \{0,1\}^{10}$, and saturation gives $\mathbf{y}^m = \mathbf{y}$. □

Consequently $\mathbb{Z}^{12} \cap \operatorname{conv}(\mathcal{Z}) = \mathcal{Z}$, and no nonzero base lies in the convex hull of the other bases. By the separation theorem for a point and a compact convex set, each nonzero $\mathbf{z} \in \mathcal{Z}$ therefore has a linear functional $\ell_{\mathbf{z}}$ on $\mathbb{R}^{12}$ with $\ell_{\mathbf{z}}(\mathbf{z}') < \ell_{\mathbf{z}}(\mathbf{z})$ for all $\mathbf{z}' \in \mathcal{Z} \setminus \{\mathbf{z}\}$.

Lemma 5.2. *The set C is a rational polytope with nonempty interior. For $\mathbf{z} \in \mathbb{Z}^{12}$, the fiber $C_{\mathbf{z}}$ equals $A_{\mathbf{z}}$ if $\mathbf{z} \in \mathcal{Z}$ and is empty otherwise. Consequently $\mu(C) = W$, and every halfspace H satisfies $\mu(C \cap H) = \sum_{\mathbf{z}\in\mathcal{Z}} \operatorname{vol}(A_{\mathbf{z}} \cap H_{\mathbf{z}})$.*

Proof. The fibers are polytopes with rational vertices, because T_j and K_j are cut from B by rational inequalities, N_j has rational vertices, and the dilation factors are rational. So C is the convex hull of finitely many rational points. It contains $\{\mathbf{0}\} \times B$, and B spans the continuous directions. The regular bases $(\mathbf{1}_{\{i\}}, \mathbf{0})$ and the near bases $(\mathbf{0}, -\mathbf{e}_1)$ and $(\mathbf{0}, -\mathbf{e}_2)$ span the integer directions, so C has nonempty interior.

Let $(\mathbf{z}, \mathbf{x}) \in C$ with $\mathbf{z}$ integer, and write $(\mathbf{z}, \mathbf{x}) = \sum_m \lambda_m (\mathbf{z}^m, \mathbf{x}^m)$ with $\lambda_m > 0$, $\sum_m \lambda_m = 1$, $\mathbf{z}^m \in \mathcal{Z}$ and $\mathbf{x}^m \in A_{\mathbf{z}^m}$. If $\mathbf{z} \neq \mathbf{0}$, then Lemma 5.1 gives $\mathbf{z}^m = \mathbf{z}$ for all m, so $\mathbf{z} \in \mathcal{Z}$ and $\mathbf{x} \in A_{\mathbf{z}}$ by convexity of $A_{\mathbf{z}}$. If $\mathbf{z} = \mathbf{0}$, then $\mathbf{x}^m/(1 + \delta\sigma(\mathbf{z}^m)) \in B$ for all m, where $1 + \delta\sigma(\mathbf{z}^m) \geq 1 - \delta > 0$. Since σ is linear, the positive weights $\lambda_m(1 + \delta\sigma(\mathbf{z}^m))$ sum to $1 + \delta\sigma(\mathbf{z}) = 1$. Thus $\mathbf{x}$ is a convex combination of the points $\mathbf{x}^m/(1 + \delta\sigma(\mathbf{z}^m))$ of B with these weights, which gives $\mathbf{x} \in B = A_{\mathbf{0}}$. The reverse inclusions are immediate, and the formulas for μ follow by summing over $\mathcal{Z}$. □

6 Cuts and the exact comparison

We now construct the cuts, starting with the three origin cuts, which contain the light fibers.

Lemma 6.1. *For $j = 1, 2, 3$ we have $\mu(C \cap H_j) = \beta + L$. Each point $(\mathbf{0}, \mathbf{x})$ with $\mathbf{x} \in B$ and each point of a light fiber lies in some H_j.*

Proof. By Lemma 5.2 we may sum over the bases, using the table of Section 4. Over the origin H_j holds $B \cap \{f_j \geq 0\}$, of volume β. If $\tau_j(\mathbf{z}) = -1$, then $\mathbf{z}$ is a light base of type j, and the condition $f_j \geq -3$ holds on its whole fiber because $|f_j| \leq 1$ on B. These fibers have total volume L. If $\mathbf{z} \neq \mathbf{0}$ and $\tau_j(\mathbf{z}) = 0$, then $\mathbf{z}$ is a paired or light base of a type $k \neq j$, or a near base of type j. Its fiber lies in $\{f_j \leq 0\}$, so H_j meets it in a subset of the null hyperplane $\{f_j = 0\}$. If $\tau_j(\mathbf{z}) \geq 1$, then $\mathbf{z}$ is heavy and the condition $f_j \geq 3$ fails on its fiber. Hence $\mu(C \cap H_j) = \beta + L$. Since $f_1 + f_2 + f_3 = 0$, each $\mathbf{x} \in B$ has $f_j(\mathbf{x}) \geq 0$ for some j, so $(\mathbf{0}, \mathbf{x}) \in H_j$. A point of a light fiber of type j lies in H_j. □

Lemma 6.2. *Let $\mathbf{z} \neq \mathbf{0}$ be a base, let ψ be a nonconstant affine function on $\mathbb{R}^5$, and let $\kappa \in \mathbb{R}$. There is a halfspace H with $H_{\mathbf{z}} = \{\psi \geq \kappa\}$ and $\mu(C \cap H) = \mathrm{vol}(A_{\mathbf{z}} \cap \{\psi \geq \kappa\})$.*

Proof. For $\rho > 0$ put $F(\mathbf{z}', \mathbf{x}') = \psi(\mathbf{x}') - \psi(\mathbf{0}) + \rho\ell_{\mathbf{z}}(\mathbf{z}')$ and $b = \kappa - \psi(\mathbf{0}) + \rho\ell_{\mathbf{z}}(\mathbf{z})$. Then F is a nonzero linear functional, and $F(\mathbf{z}, \mathbf{x}') \geq b$ if and only if $\psi(\mathbf{x}') \geq \kappa$. Since $\mathcal{Z}$ is finite, the fibers are compact and $\ell_{\mathbf{z}}(\mathbf{z}') < \ell_{\mathbf{z}}(\mathbf{z})$ for $\mathbf{z}' \neq \mathbf{z}$, we have $F < b$ on every fiber over $\mathbf{z}' \neq \mathbf{z}$ once ρ is large. Lemma 5.2 gives the mass of $H = \{F \geq b\}$. □

A regular or near fiber is a simplex of volume h^5. Some barycentric coordinate of a point in it is at least $1/6$. That coordinate is affine and nonconstant, and the cap of the simplex where it is at least $1/6$ is a homothetic copy with ratio $5/6$, so Lemma 6.2 gives a cut of mass $M = (5h/6)^5$ containing the point.

A paired fiber is not a simplex, and we cover it with the coordinates ν_i of U. For U alone we would use the thresholds $1/6$. The slab adds unequally to the six resulting parts, and we balance them with thresholds close to $1/6$. Fix positive rational numbers $t_0, \ldots, t_5$ with $\sum_i t_i = 1$ and $1/7 < t_i < 1/5$. Let $\mathbf{z}$ be a paired base of type j and let $\mathbf{x} \in (1 + \delta)K_j$. By rotation we compute with $j = 1$. Since $\sum_i \nu_i = \sum_i t_i = 1$ on all of $\mathbb{R}^5$, there is an index i with $\nu_i(\mathbf{x}/(1 + \delta)) \geq t_i$. The function $\mathbf{x}' \mapsto \nu_i(\mathbf{x}'/(1 + \delta))$ is affine and nonconstant, so Lemma 6.2 with $\kappa = t_i$ gives a cut containing $(\mathbf{z}, \mathbf{x})$ of mass $(1 + \delta)^5 V_i$, where $V_i = \mathrm{vol}(K_1 \cap \{\nu_i \geq t_i\})$.

Lemma 6.3. *We have $V_2 = \theta(1 - t_2)^5$, and for $i \neq 2$,*

$$V_i = \theta(1 - t_i)^5 + \phi \frac{(1 - t_i)^5 - \big(1 - t_i - (1 - q_i)\varepsilon\big)^5}{1 - q_i}.$$

Proof. We split K_1 into U and $P \cap \{p \le \varepsilon\}$, which overlap in a null set. In U the set $\{\nu_i \ge t_i\}$ is a homothetic copy of U with ratio $1 - t_i$, of volume $\theta(1 - t_i)^5$. On P with $i = 2$ we have $\nu_2 = q_2 p \le 0 < t_2$, so this part is empty. Let $i \ne 2$. For $0 \le p \le \varepsilon$, the bounds $0 \le q_i < 1$, $\varepsilon < 1/1000$ and $1/7 < t_i < 1/5$ give $t_i - q_i p > 0$, $1 - t_i - (1 - q_i)p > 0$ and $1 - q_i > 0$. The affine chart with coordinates p and p_k, $k \ne i, 2$, sends P onto the standard simplex of coordinate Lebesgue volume $1/120$, so volume in our units on P is 120ϕ times coordinate Lebesgue measure. Substituting $p_i = 1 - p - \sum_{k \ne i,2} p_k$ turns the condition $\nu_i = p_i + q_i p \ge t_i$ into $\sum_{k \ne i,2} p_k \le 1 - t_i - (1 - q_i)p$, and it forces $p_i \ge t_i - q_i p > 0$. So for each p the four remaining coordinates range over a simplex of coordinate Lebesgue volume $[1 - t_i - (1 - q_i)p]^4/24$. Multiplying by 120ϕ and integrating over p, the contribution of P is

$$5\phi \int_0^{\varepsilon} \big[1 - t_i - (1 - q_i)p\big]^4 \, dp = \phi \, \frac{(1 - t_i)^5 - \big(1 - t_i - (1 - q_i)\varepsilon\big)^5}{1 - q_i}. \qquad \square$$

The cuts above reduce the theorem to three strict comparisons of explicit rational numbers,

$$\beta + L < M, \qquad (1 + \delta)^5 V_i < M \quad (i = 0, \dots, 5), \qquad 4096 h^5 < W. \tag{3}$$

The first bounds the origin cuts, the second bounds the cuts on the paired fibers, and the third is the target (1). We take

$$a = \frac{663}{1000}, \qquad \delta = \frac{664201}{10^6}, \qquad \varepsilon = \frac{13}{20000}, \qquad h = \frac{1111946415}{10^9},$$
$$(t_0, \dots, t_5) = 10^{-9}(166735371,\ 166735294,\ 166324902,\ 166734811,\ 166734811,\ 166734811).$$

They satisfy all the conditions imposed so far, namely $1/2 < a < 1$, $0 < \delta < 1$, $0 < \varepsilon < 1/1000$, $h > 0$, $2h < 1 + 2\delta$, $\sum_i t_i = 1$ and $1/7 < t_i < 1/5$. Exact evaluation of the formulas of Sections 4 and 6 gives $M - \beta - L > 10^{-6}$, $M - \max_i (1 + \delta)^5 V_i > 10^{-9}$ and $W - 4096 h^5 > 4 \cdot 10^{-5}$, which proves (3). Appendix A lists the values. In (2), the gain $3L$ exceeds the deficit $h^5 - 1$ of the origin fiber by $2.8 \cdot 10^{-4}$, and the 756 paired fibers, each short of h^5 by $3.1 \cdot 10^{-7}$, use up $2.4 \cdot 10^{-4}$ of this.

Proof of Theorem 1.1. Lemma 5.2 gives the stated properties of C and $\mu(C) = W > 0$ in our units. A point $(\mathbf{z}, \mathbf{x}) \in \mathbb{Z}^{12} \times \mathbb{R}^5$ either lies outside C or has $\mathbf{z} \in \mathcal{Z}$ and $\mathbf{x} \in A_{\mathbf{z}}$. If $(\mathbf{z}, \mathbf{x}) \notin C$, strict separation gives a halfspace that contains the point and misses C, so it has mass 0. If $\mathbf{z} = \mathbf{0}$ or $\mathbf{z}$ is light, Lemma 6.1 gives a containing H_j of mass $\beta + L < M$. If $\mathbf{z}$ is regular or near, the cap of the simplex gives mass M. If $\mathbf{z}$ is paired, the cover by the ν_i gives mass $(1 + \delta)^5 V_i < M$. So every point of $\mathbb{Z}^{12} \times \mathbb{R}^5$ lies in a halfspace of mass at most M and has depth at most $r = M/W = (5/6)^5 h^5 / W$, and $W > 4096 h^5$ gives $r < (5/6)^5 / 2^{12}$. $\square$

A Lean statement and exact values

The repository `https://github.com/Hongyu-Cheng/weak-oertel-conjecture` holds the Lean 4 project in its folder `lean` and the exact checker. With Python 3 and `elan` installed, the following commands, run from the root of the repository, rerun both.

```
python3 research/sharp-dimension-17/check.py
cd lean && lake exe cache get && lake build && lake env lean CheckAxioms.lean
```

The last command fails unless the final theorem and the other results it lists, among them the lemma `integer_projection_census` that $\mathbb{Z}^{12} \cap \mathrm{conv}(\mathcal{Z}) \subseteq \mathcal{Z}$, depend only on the axioms `propext`, `Classical.choice` and `Quot.sound`. In the file `lean/SharpOC/Selector17Counterexample.lean`, the

final theorem `SharpOC.Selector17.sharp_oc_counterexample` states that there are a compact convex set C with nonempty interior in `(Fin 10 ⊕ Fin 2 → ℝ) × (Fin 5 → ℝ)`, with $\mathbf{0} \in C$ and $\mu(C)$ positive and finite, and a real number r with $0 < r < (5/6)^5/2^{12}$ such that for all $\mathbf{z} \in \mathbb{Z}^{12}$ and $\mathbf{x} \in \mathbb{R}^5$ there are a nonzero continuous linear functional F and $b \in \mathbb{R}$ with $b \le F(\mathbf{z}, \mathbf{x})$ and $\mu(C \cap \{F \ge b\})/\mu(C) \le r$. Here μ is `mixedIntegerMass`, the extended nonnegative sum over $\mathbb{Z}^{12}$ of the Lebesgue volumes of the fibers, as in Section 1. This is the bound of Theorem 1.1. Rationality of the polytope is not part of the Lean statement and is established by Lemma 5.2.

The checker generates the 4852 bases and their counts and asserts, in exact rational arithmetic, the conditions on the parameters and the three margins of Section 6, which imply (3). It does not show that the bases exhaust the integer points of C, which is proved in Section 5 and in Lean. The following floors come from direct exact evaluation of the formulas of Sections 4 and 6.

Mass	$\lfloor 10^{12} \cdot \text{mass} \rfloor$	Margin	$\lfloor 10^{12} \cdot \text{margin} \rfloor$
h^5	1699884034250	$M - \beta - L$	1262439
M	683145268393	$M - (1+\delta)^5 V_0$	7542
β	449755355574	$M - (1+\delta)^5 V_1$	6288
L	233388650379	$M - (1+\delta)^5 V_2$	1211
$3L$	700165951139	$M - (1+\delta)^5 V_i,\ i = 3, 4, 5$	7869
$(1+\delta)^5\theta$	1696402554056	$h^5 - (1+\delta)^5 V$	312774
$(1+\delta)^5 V$	1699883721475	$1 + 3L - h^5$	281916889
W	6962725049747587	$W - 4096h^5$	45459473

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